					Course Code:A2610001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		ENGLISH for TECHNICAL COMMUNICATION			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Identify any four expressions for asking for permissions.	**1**	**2**	**2**
	b).	Complete the sentences with the appropriate words as directed in the brackets. i. The students are interested ___ learning new technologies. (Preposition) ii. The success of _____ project depends on effective teamwork. (Article)	**2**	**3**	**2**
	c).	Construct one example sentence for any two of the following idiomatic expressions. i. A piece of cake ii. Break the ice iii. Once in a blue moon iv. Spill the beans	**3**	**3**	**2**
	d).	Write about the use of skimming technique in reading the texts.	**4**	**3**	**2**
	e).	Dos and don'ts of social media messaging.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Develop an essay on the strategies that can help a person sound smarter, more confident, and articulate when communicating.	**1**	**3**	**5**
	b).	Write a dialogue between a student and a librarian about borrowing a reference book and renewing the library membership.	**1**	**3**	**5**
		OR			
3.		Construct a paragraph on "The Importance of Time Management" in about 100–120 words and list out the cohesive devices used in the paragraph.	**1**	**3**	**10**
		UNIT-2			
4.	**a).**	Explain how eco-conscious negotiation strategies can be applied to address an environmental issue and promote a sustainable future.	**2**	**3**	**5**

	b).	Employ appropriate verb forms to fill in the blanks. i. By next year, she ___ (finish) her degree. ii. The train ___ (arrive) at the station just now. iii. I ___ (study) English for three years. iv. When I called him, he ___ (sleep). v. Listen! Someone ___ (play) the guitar.	**2**	**3**	**5**
		OR			
5.		Apply appropriate grammatical rules and correct any **TEN** of the following sentences. i. Each of the engineers are responsible for testing the software. ii. The equipment in the laboratory were not functioning properly. iii. He discussed about the advantages of artificial intelligence. iv. The results of the experiment shows a significant improvement. v. The engineer suggested to use a different algorithm for the project. vi. If the system will fail, the backup server will be activated. vii. He is resembling his father. viii. The film show began before we reached the theatre. ix. Neither the project manager nor the team members was aware of the technical issue. x. She is an university student pursuing engineering. xi. Rahul went to school despite of having a fever. xii. I prefer chocolate milkshake than vanilla ice cream.	**2**	**3**	**10**
		UNIT-3			
6.	**a).**	Illustrate the appropriate ethical guidelines to address a challenge arising from the use of artificial intelligence or other disruptive technologies.	**3**	**3**	**5**
	b).	Construct sentences by substituting the given phrases with suitable one-word for any **FIVE** of the following. i. One who loves mankind. ii. One who cannot be defeated iii. Hobby of collecting coins. iv. Something that cannot be avoided v. A speech made without preparation vi. A person who spends money wastefully vii. A person who designs and develops software viii. A person who loves and collects	**3**	**3**	**5**
		OR			
7.	**a).**	Construct a formal letter to the Editor of a newspaper expressing your concern about the increasing use of plastic in your city. Apply appropriate letter-writing conventions and suggest possible solutions.	**3**	**3**	**10**

		UNIT-4			
8.	**a).**	Develop an essay on how innovative business practices can be applied to address challenges faced by modern startups.	**4**	**3**	**5**
	b).	Analyze the following hints to construct a formal email by applying appropriate writing techniques, tone, and structure. **Student ---> HR Manager:** New training/internship program -latest technology- useful – my career-want to be included- participate keenly-learn and implement- benefit of the company.	**4**	**3**	**5**
		OR			
9.	**a).**	Analyze the given passage and write answers to the following questions. Technology has transformed the way students learn and access information. Digital platforms, online courses, and educational applications allow learners to study at their own pace and explore subjects beyond the classroom. Engineering students, in particular, can use simulations, virtual laboratories, and collaborative tools to understand complex concepts more effectively. However, technology should complement traditional learning rather than completely replace it. Excessive dependence on digital resources may reduce face-to-face interaction and critical thinking. Therefore, students should use technology responsibly and balance digital learning with classroom discussions, practical activities, and independent study. Questions i. How has technology transformed the learning process? ii. What are the advantages of digital platforms and online courses? iii. How can engineering students benefit from simulations and virtual laboratories? iv. Why should technology not completely replace traditional learning? v. What are the possible disadvantages of excessive dependence on digital resources? vi. What does the word “complement” mean in the context of the passage? vii. Find a word from the passage that means “working together.” viii. Find a word from the passage that means “too much or more than necessary.”	**4**	**3**	**10**

		ix. What three activities does the passage recommend balancing with digital learning? x. What is the central idea of the passage?			
		UNIT-5			
10.	**a).**	Demonstrate how core values can be integrated into ethical decision-making and socially responsible leadership.	**5**	**3**	**5**
	b).	Construct the following sentences as directed in the brackets. i. She has written many novels. (Add Question Tag) ii. The report was prepared by the committee. (Convert it into Active Voice) iii. He worked hard to pass the examination. (Convert it into a Complex Sentence) iv. Very few engineering colleges in India are as reputed as IIT Madras. (Convert it into the Superlative Degree) v. Smrithi is the cleverest girl in the class. (Change it into Comparative Degree)	**5**	**3**	**5**
		OR			
11.	**a).**	Develop the content to be used in a blog on 'Social Media'.	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2610002
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		LINEAR ALGEBRA & CALCULUS			
		(Common to All Branches of Engineering)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	KL	M
1.	(a)	Find the value of k such that rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2	1	3	2
	(b)	Find the singular values of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	2	3	2
	(c)	Find the value of c in [0, 4] for $f(x) = x(x-2)(x-3)$ by using Lagrange's mean value theorem.	3	3	2
	(d)	Evaluate $\underset{\substack{x\to\infty \\ y\to 2}}{Lt}\left(\frac{xy+1}{x^2+2y^2}\right)$	4	3	2
	(e)	Evaluate $\int_0^2 \int_0^4 (x^2+y^2)\,dxdy$	5	3	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	(a)	Determine the rank of the matrix $\begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$ by reducing it to echelon form.	1	3	5
	(b)	Find the inverse of the matrix $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ by using Gauss – Jordan method.	1	3	5
		OR			
3.	(a)	For what values of a and b do the equations $x+2y+3z=6$, $x+3y+5z=9$ and $2x+5y+az=b$ have (i) No solution (ii) An unique solution (ii) Infinitely many solutions	1	3	5
	(b)	Solve the equations $20x+y-2z=17$, $3x+20y-z=-18$and $2x-3y+20z=25$ by Gauss-Seidel iteration method.	1	3	5
		UNIT-2			

4	Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$		2	3	10
	OR				
5.	Reduce $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ to sum of squares by linear transformation and hence find the matrix of the transformation, Nature, Signature and index of the quadratic form.		2	3	10
		UNIT-3			
6.	(a)	Verify Rolle's theorem for $\frac{\sin x}{e^x}$ in $(0, \pi)$.	3	3	5
	(b)	Expand $f(x) = \cos x$by Maclaurin's theorem with Lagrange's form of remainder.	3	3	5
		OR			
7.	(a)	Find the appropriate value of c used in Cauchy's Mean-Value theorem for $f(x) = \ln x$ in [1, e]	3	3	5
	(b)	Expand $\tan^{-1}x$ in powers of $(x-1)$ upto fourth degree term.	3	3	5
		UNIT-4			
8.	(a)	If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$	4	3	5
	(b)	Expand $f(x,y) = e^y \log(1+x)$ in powers of x and y using Maclaurin's series.	4	3	5
		OR			
9.	(a)	If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	4	3	5
	(b)	Discuss the maxima and minima of $f(x,y) = x^3y^2(1-x-y)$	4	3	5
		UNIT-5			
10.	(a)	Evaluate $\int\int xydxdy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$ by changing it in to polar coordinates	5	3	5
	(b)	By the change of order of integration evaluate $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dydx$.	5	3	5
		OR			
11.		Evaluate $\iiint_V \frac{dxdydz}{x^2+y^2+z^2}$ where V is the volume of the sphere $x^2 + y^2 + z^2 = a^2$	5	3	10

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610004
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		PHYSICS FOR ADVANCED COMPUTING			
		(Common to CSG, CSBS & CSIT)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE** Question from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a)**	Explain conditions for constructive and destructive interference.	1	2	2
	b)	Distinguish between Spontaneous and Stimulated emissions.	2	3	2
	c)	State Heisenberg's uncertainty principle.	3	2	2
	d)	Explain drift and diffusion currents in semiconductors	4	2	2
	e)	State and Explain Meissner effect.	5	2	2
					5 x 10 = 50 Marks
		UNIT-1	**CO**	**KL**	**M**
2.		With relevant theory, determine the wavelength of monochromatic light and refractive index of a liquid medium in Newton's rings experiment.	1	3	10
		OR			
	a)	Distinguish Fresnel and Fraunhofer diffraction.	1	3	5
	b)	Derive the expression for thickness of a half wave plate and a quarter wave plate.	1	3	5
		UNIT-2			
4.	**a)**	Explain population inversion in lasing system.	2	3	5
	b)	Discuss the construction and working of Ruby Laser.	2	3	5
		OR			
5.	**a)**	Explain the terms numerical aperture and acceptance angle of an optical fiber.	2	3	5
	b)	Describe mode of propagation of light in graded index optical fiber with neat diagram.	2	3	5
		UNIT-3			
6.		Derive the expression for electrical conductivity in metals based on Quantum free electron theory.	3	3	5
		OR			
7.	**a)**	Describe the physical significance of wave function and mention its	3	3	5

		properties.			
	b)	Deduce Schrodinger's independent wave equation for a free particle.	3	4	5
		UNIT-4			
8.		Derive the expression for the density of charge carriers in the conduction band of an intrinsic semiconductor.	4	3	10
		OR			
9.	**a)**	State and explain Hall effect. Obtain an expression for the Hall Coefficient of a material.	4	3	5
	b)	Analyze the properties of Dia, Para and Ferro magnetic materials using magnetic dipole moment.	4	4	5
		UNIT-5			
10.	**a)**	Discuss the important properties of materials at low temperatures.	5	3	5
	b)	Compare liquid Nitrogen and liquid Helium with respect to their properties and applications.	5	4	5
		OR			
11.	**a)**	Distinguish between Type I and Type II superconductors.	5	3	5
	b)	Explain BCS theory of superconductors.	5	3	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610006
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		PROBLEM SOLVING SKILLS USING C			
		(Common to AIDS & CSBS)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Explain the rules for naming identifiers in C with suitable examples.	**1**	**2**	**2**
	b).	Differentiate between break and continue statements.	**2**	**2**	**2**
	c).	Explain the concept of a union in C with a suitable example.	**3**	**2**	**2**
	d).	Explain the concept of a pointer in C with a suitable example.	**4**	**2**	**2**
	e).	Explain how the #define directive is used to create a macro with an example.	**5**	**2**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Explain the concept of Computational Thinking and its main components with a suitable example.	**1**	**2**	**5**
	b).	Explain the different data types in C with suitable examples.	**1**	**2**	**5**
		OR			
3.	**a).**	Explain the methods used to convert a decimal number into binary, octal, and hexadecimal number systems with suitable examples.	**1**	**2**	**5**
	b).	Explain the different types of operators in C with suitable examples.	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Implement a C program using the switch statement to create a menu-driven calculator that performs addition, subtraction, multiplication, and division.	**2**	**3**	**5**
	b).	Explain the different looping statements in C with suitable examples.	**2**	**2**	**5**
		OR			
5.	**a).**	Explain the necessary condition for matrix multiplication and write a C program to multiply two matrices.	**2**	**2**	**5**
	b).	Implement a C program to sort a given array of elements in ascending order using the Bubble Sort technique.	**2**	**3**	**5**
		UNIT-3			

6.	**a).**	Explain the commonly used string handling functions in C with their syntax and suitable examples.	**3**	**2**	**5**
	b).	Implement a C program using a structure named Employee to store employee ID, name, and salary, and display the employee details.	**3**	**3**	**5**
		OR			
7.	**a).**	Implement a C program to read a string and count the number of vowels, consonants, digits, and special characters in the given string.	**3**	**3**	**5**
	b).	Explain typedef and enumerated data types in C with their syntax and suitable examples.	**3**	**2**	**5**
		UNIT-4			
8.	**a).**	Explain pointers in C, including pointer declaration, initialization, and accessing addresses and values, with suitable examples.	**4**	**2**	**5**
	b).	Illustrate the parameter passing techniques in C with suitable examples.	**4**	**3**	**5**
		OR			
9.	**a).**	Explain dynamic memory allocation in C using malloc() and calloc() with suitable examples.	**4**	**2**	**5**
	b).	Implement a C program to calculate the factorial of a given number n using recursion.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain preprocessor directives and macros in C with suitable examples.	**5**	**2**	**5**
	b).	Implement a C program using file handling functions to create, store, and display student data in a file.	**5**	**3**	**5**
		OR			
11.	**a).**	Explain the commonly used file handling functions in C with their syntax and suitable examples.	**5**	**2**	**5**
	b).	Implement a C program to copy the contents of a source file into a destination file using file handling functions.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2620001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Find the integrating factor of $\frac{dy}{dx} - \frac{3y}{x} = x$	**1**	**3**	**2**
	b).	Solve$(D^2 - 4D + 4)y = 0$	**2**	**3**	**2**
	c).	Obtain a P.D.E from $z = ax + by + c$	**3**	**2**	**2**
	d).	Determine $\nabla\emptyset$, where $\emptyset = \sqrt{x^2 + y^2 + z^2}$	**4**	**3**	**2**
	e).	Find div $\bar{F}$, where $\bar{F} = x\bar{\imath} + y\bar{\jmath} + z\bar{k}$	**5**	**3**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Solve $y \log y \, dx + (x - \log y)dy = 0$	**1**	**3**	**5**
	b).	Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$	**1**	**3**	**5**
		OR			
3.	**a).**	If the air is maintained at 30ºC and the temperature of the body cools from 100ºC to 70º C in 15 minutes, determine the time at which the temperature of the body will be 40º C.	**1**	**3**	**5**
	b).	Find the orthogonal trajectories of the family of curves $r^2 = a^2 Sin2\theta$	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Solve $(D-2)^2 y = 8(e^{2x} + x^2)$	**2**	**3**	**5**
	b).	Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + y = \tan x$	**2**	**3**	**5**
		OR			
5.	**a).**	Solve simultaneous equations $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that $x = 2$ and $y=0$ when $t = 0$	**2**	**3**	**5**
	b).	The damped LCR circuit is governed by the differential equation	**2**	**3**	**5**

		$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$ where L, C, R are positive constants. Find the critical resistance.			
		UNIT-3			
6.	**a).**	Form the partial differential equation by eliminating arbitrary function from $f(x^2 + y^2 + z^2, z^2 - 2xy) = 0$.	**3**	**3**	**5**
	b).	Solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$	**3**	**3**	**5**
		OR			
7.		Solve $\left(D^2 - 3DD' + 2D'^2\right) z = e^{x+2y} + \mathrm{Sin}(x+2y)$	**3**	**3**	**10**
		UNIT-4			
8.	**a).**	Find the directional derivative of $f = xy+yz+zx$ in the direction of vector $\bar{i} + 2\bar{j} + 2\bar{k}$ at the point (1,2,0).	**4**	**3**	**5**
	b).	If $\overrightarrow{F} = 3x^2\overrightarrow{\imath} + 5xy^2\overrightarrow{j} + 5xyz^3\vec{k}$,then find $\nabla(\nabla.\vec{F})$	**4**	**3**	**5**
		OR			
9.	**a).**	Prove that $(y^2 - z^2 + 3yz - 2x)\vec{\imath} + (3xz + 2xy)\overrightarrow{j}$ + (3xy-2xz+2z)$\vec{k}$ is solenoidal.	**4**	**3**	**5**
	b).	Find the angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point $(1,-2,1)$	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Find the work done by the force field$\vec{F}$= $(2x + y^2)\vec{\imath} + xz\overrightarrow{j} + xyz\vec{k}$in moving a particle along a straight line from (0,0,0) to (2,1,3).	**5**	**3**	**5**
	b).	Evaluate $\oint_c \{(x^2y\,dx + x^2dy\}$ where C is a circle $x^2 + y^2 = a^2$ by using Green's theorem.	**5**	**3**	**5**
		OR			
11.		Verify Gauss – divergence theorem for$\vec{F}$ = $2xz\overrightarrow{\imath}$ +yz$\vec{j} + z^2\vec{k}$ where S is an upper half surface of the sphere$x^2 + y^2 + z^2 = a^2$	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620003
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		APPLIED CHEMISTRY FOR EMERGING TECHNOLOGIES			
		(Common to CSBBS, CSD & CSIT)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Apply molecular orbital theory to explain the paramagnetic nature of O_2.	**1**	**3**	**2**
	b).	Distinguish between primary and secondary batteries with suitable examples.	**2**	**2**	**2**
	c).	Determine the wavelength range of visible radiation from the electromagnetic spectrum.	**3**	**3**	**2**
	d).	Use the principle of photolithography to explain pattern formation on a semiconductor wafer.	**4**	**3**	**2**
	e).	Explain the significance of nanoscale dimensions in nanochemistry.	**5**	**2**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Utilize Planck's quantum theory to explain black-body radiation and demonstrate how it resolves the limitations of classical theory.	**1**	**3**	**5**
	b).	Solve the time-independent Schrödinger wave equation for a particle confined in a one-dimensional box and derive the expression for its allowed energy levels.	**1**	**3**	**5**
		OR			
3.	**a).**	Construct molecular orbital energy-level diagrams for HF and CO and apply molecular orbital theory to explain their bonding characteristics.	**1**	**3**	**5**
	b).	Sketch the π-molecular orbital energy-level diagrams of benzene and apply the concept of electron delocalization to explain their bonding.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Construct a calomel electrode and explain how it can be used as a reference electrode to determine the electrode potential of an unknown electrode.	**2**	**3**	**5**

	b).	Use the concepts of battery capacity, power density, cell balancing, and cycle life to evaluate the performance requirements of a battery for an electric vehicle.	**2**	**3**	**5**
		OR			
5.	**a).**	Illustrate the working of zinc–air and lithium primary batteries and relate their characteristics to identify suitable applications.	**2**	**3**	**5**
	b).	Model a lithium-ion battery and apply the electrode reactions to explain its charging and discharging processes.	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	Sketch a block diagram of an electrochemical sensor and apply its working principle to explain the operation of potentiometric sensors.	**3**	**3**	**5**
	b).	Apply the colorimetric method to explain the estimation of iron in a water sample, including the principle, instrumentation, and procedure.	**3**	**3**	**5**
		OR			
7.	**a).**	Prepare a labelled block diagram of a UV-Visible spectrophotometer and apply the functions of its components to explain its working principle.	**3**	**3**	**5**
	b).	Employ the principle of IR spectroscopy to explain the stretching and bending vibrational modes of molecules, with suitable examples.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Construct a schematic representation of the zone-refining process and apply its principle to explain the purification of semiconductor-grade silicon.	**4**	**3**	**5**
	b).	Apply the conduction mechanism of polyacetylene to explain its electrical conductivity and identify its applications in electronic devices.	**4**	**3**	**5**
		OR			
9.	**a).**	Model a labelled diagram of a solar photovoltaic cell and apply its working principle to explain the conversion of solar energy into electrical energy.	**4**	**3**	**5**
	b).	Relate the characteristics of Type-I and Type-II superconductors to distinguish their behaviour in magnetic fields and identify suitable applications.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Interpret the structural and electronic properties of carbon nanotubes and graphene to explain their applications in nanoelectronics and sensing.	**5**	**3**	**5**
	b).	Illustrate the arc-discharge method for the synthesis of carbon nanomaterials and identify their applications in emerging technologies.	**5**	**3**	**5**
		OR			

11.	**a).**	Demonstrate the applications of nanomaterials in nanoelectronics, interconnects, sensors, memory devices.	**5**	**3**	**5**
	b).	Utilize the basic concepts of spintronics to explain the roles of magnetic nanoparticles and ferromagnetic materials in spin-based electronic devices.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620004
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		DATA STRUCTURES USING C			
		(Common to AIDS, AIML, CIC, CSBS, CSD, CSE, CSIT, IT)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Explain the concept of an Abstract Data Type (ADT) with a suitable example.	**1**	**2**	**2**
	b).	Distinguish between Singly Linked List and Doubly Linked List.	**2**	**2**	**2**
	c).	Explain the condition that indicates stack underflow.	**3**	**2**	**2**
	d).	Differentiate between the height and depth of a tree.	**4**	**2**	**2**
	e).	Describe collision in hashing and state why it occurs.	**5**	**2**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Explain classification of data structures with examples.	**1**	**2**	**5**
	b).	Use Binary Search to find **75** in: 10, 25, 35, 45, 55, 65, 75, 85, 95. Show the sequence of comparisons.	**1**	**3**	**5**
		OR			
3.	**a).**	Explain time complexity and space complexity of an algorithm with suitable examples.	**1**	**2**	**5**
	b).	Describe the Quick sort algorithm by sorting the following collection of numbers as an example, 42, 7, 52, 57, 62, 37, 32, 27, 22.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Explain the different Dynamic Memory Allocation functions in C	**2**	**2**	**5**
	b).	Apply the basic operations of a Circular Linked List and illustrate them with a suitable example.	**2**	**3**	**5**
		OR			
5.	**a).**	Apply recursion to generate the Fibonacci sequence for n = 6. Show the sequence of recursive calls	**2**	**3**	**5**
	b).	Explain the algorithm for inserting a node at the beginning and at the end of a Singly Linked List.	**2**	**2**	**5**

		UNIT-3			
6.	**a).**	Apply the PUSH, POP, PEEK, and isEmpty operations on a stack using a suitable example. Show the value of TOP after each operation.	**3**	**3**	**5**
	b).	Implement a queue using a linked list and illustrate the enqueue and dequeue operations with suitable diagrams.	**3**	**3**	**5**
		OR			
7.	**a).**	Evaluate the given postfix expression using a stack and show the stack after each operation.	**3**	**3**	**5**
	b).	Apply operations on a circular queue and demonstrate how vacant positions are reused.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Explain the different tree traversal techniques with a suitable example.	**4**	**2**	**5**
	b).	Apply the deletion operation in a Binary Search Tree (BST) using a suitable algorithm and example.	**4**	**3**	**5**
		OR			
9.	**a).**	Explain the different methods of representing a binary tree.	**4**	**2**	**5**
	b).	Apply Min Heap and Max Heap concepts to construct both heaps using the elements 20, 10, 30, 5, 40.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Explain the Depth First Search (DFS) traversal technique with a suitable algorithm and example.	**5**	**2**	**5**
	b).	Explain Prim's algorithm for finding the Minimum Spanning Tree (MST) of a weighted graph with a suitable example.	**5**	**2**	**5**
		OR			
11.		Explain the concept of open addressing in hashing and explain the different collision resolution techniques used in open addressing with suitable examples.	**5**	**2**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620006
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		FUNDAMENTALS OF PYTHON PROGRAMMING			
		(Common to CSBS, CSD & CSIT)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	List any four basic data types available in Python.	**1**	**2**	**2**
	b).	difference between string indexing and slicing	**2**	**2**	**2**
	c).	List higher order functions in python.	**3**	**2**	**2**
	d).	Importance of the __init__() method.	**4**	**2**	**2**
	e).	Explain the importance of encapsulation in OOP.	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Write a Python program to accept a student's marks, calculate the average, and display the grade using if, if-else, and chained conditional statements.	**1**	**3**	**5**
	b).	Develop a Python program that uses break, continue, and pass to control the execution of a loop.	**1**	**3**	**5**
		OR			
3.	**a).**	Explain the special operators in Python, namely membership operators and identity operators, with suitable examples..	**1**	**2**	**5**
	b.)	Explain how an algorithm and flowchart can be translated into a Python program.	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Develop a Python program that accepts a string and performs various operations such as finding its length, reversing it, searching for a substring, counting characters, and changing its case. Explain the string methods used.	**2**	**3**	**10**
		OR			
5.	**a).**	Write a Python program to create a list and **apply** list methods such as append(), insert(), remove(), pop(), and sort() to manipulate the list.	**2**	**3**	**5**
	b).	Write a Python program to count the frequency of each word in a given sentence using a dictionary.	**2**	**3**	**5**

		UNIT-3			
6.	**a).**	Define a module and a package in Python. Explain the steps involved in creating and using a user-defined package with a suitable example.	**3**	**2**	**5**
	b).	Explain the types of user-defined functions and arguments in Python with examples.	**3**	**2**	**5**
		OR			
7.	**a).**	Explain the use of map()**,** filter()**, and** reduce() functions with suitable examples.	**3**	**2**	**5**
	b).	Write a Python program using recursion to find the Fibonacci series up to **n** terms.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Write a Python program to create a class containing public and private data members. Demonstrate how they are accessed through appropriate methods.	**4**	**3**	**5**
	b).	Discuss encapsulation, abstraction, and polymorphism in detail with suitable examples	**4**	**2**	**5**
		OR			
9.	**a).**	Explain inheritance in Python. Discuss the different types of inheritance with suitable examples.	**4**	**2**	**10**
		UNIT-5			
10.	**a).**	Explain the file handling methods used for reading and writing files in Python.	**5**	**3**	**5**
	b).	Define exception handling. Explain the purpose of try, except, and finally blocks with suitable examples.	**5**	**3**	**5**
		OR			
11.	**a).**	Write a Python program to read a text file and count the number of lines, words, and characters. Write the results into another file.	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620008
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		Sustainable Development: Principles and Practices			
		(Common to CSBS, CSIT & CSD)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Define sustainable development. What is meant by intergenerational equity?	1	1	2
	b).	Name any four of the 17 SDGs and state one target of any one of them.	2	1	2
	c).	Differentiate between climate change mitigation and adaptation.	3	2	2
	d).	How does gender equality contribute to sustainable development?	4	2	2
	e).	What is the localization of SDGs?	5	1	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	a).	Explain the environmental, social and economic dimensions of sustainability with a neat sketch.	1	2	5
	b).	Discuss the principles of sustainable development and its significance for quality of life, from global, national and local perspectives.	1	2	5
		OR			
3.	a).	Trace the evolution of sustainable development from the Brundtland Commission to Agenda 2030. Explain the transition from MDGs to SDGs.	1	2	10
		UNIT-2			
4.	a).	Describe the 17 SDGs with their focus areas, and group them under the people, planet, prosperity, peace and partnership themes.	2	2	10
		OR			
5.	a).	Explain the interlinkages between different Sustainable Development Goals with suitable examples.	2	2	5
	b).	Interpret SDG 9 and SDG 11 with their major targets, and show how they connect to other goals	2	3	5

		UNIT-3			
6.	a).	Discuss the causes and impacts of climate change. Suggest mitigation and adaptation measures for a city.	3	3	10
		OR			
7.	a).	Explain the circular economy approach to waste management and sustainable production and consumption	3	3	5
	b).	Mention one local environmental problem and suggest a sustainable solution.	3	3	5
		UNIT-4			
8.	a).	Explain how poverty, hunger, health and education are interlinked. Discuss strategies to achieve SDGs 1, 2, 3 and 4 together.	4	2	10
		OR			
9.	a).	Discuss how gender equality, social inclusion and reduced inequalities support inclusive development	4	2	5
	b).	Explain the role of technology and innovation in decent work and inclusive economic growth	4	2	5
		UNIT-5			
10.	a).	Explain how SDG progress is measured using targets and indicators. Describe the roles of government, institutions and governance in SDG implementation in India.	5	2	10
		OR			
11.	a).	Explain the localization of SDGs and the role of higher education institutions and students in community engagement.	5	2	5
	b).	Prepare a simple SDG-based action plan for a village issue such as waste, water or energy.	5	3	5

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